

3.58:

```
long decode2(long x, long y, long z)
{
    y -= z;
    x *= y;
    long temp = y;
    temp=temp << 63 >> 63;
    return x ^ temp;
}
```

3.60:

```
long loop(long x, int n)
{
    long result = 0;
    long mask;
    for (mask = 1; mask != 0; mask = (mask << n))
    {
        result |= (mask & x);
    }
    return result;
}
```

```
A:x:%rdi,%r8;
n:%esi, (%ecx,%cl);
result: %rax;
mask: (%edx, %rdx);
```

B:result 初始化为 0; mask 初始化为 1;

C: mask!=0;

D: mask = (mask << n);

E:result |= (mask & x);

F:如上。

3.63:

```
long switch_prob(long x, long n)
{
    long result = x;
    switch (n)
    {
        case 60:
        case 62:
            result = 8 * x;
            break;
        case 63:
            result >>= 3;
    }
}
```

```
        break;
    case 64:
        result <<= 4;
        result -= x;
        x = result;
    case 65:
        x *= x;
    default:
        result = 75 + x;

}
return result;
}
```